

Two-Model Local AI Chatbot

Complete configuration and implementation guide for RTX 4060 8 GB

Architecture: Qwen3 8B → DeepSeek-R1 8B verifier → correction/final answer, with optional web search and RAG.

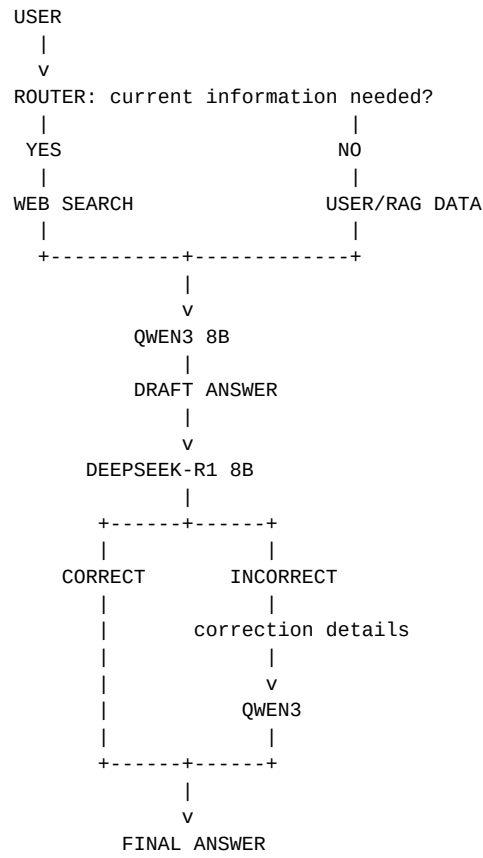
1. Project Objective

Build a local-first chatbot that answers English questions using Qwen3 8B as the primary answer generator and DeepSeek-R1 Distill 8B as a verifier. For questions requiring current information, the application retrieves web information first, passes the retrieved context to Qwen, and then asks DeepSeek to verify relevance, accuracy, freshness, and English quality.

2. Target Hardware

Component	Target
GPU	NVIDIA RTX 4060 8 GB (your GPU)
System RAM	16 GB minimum; 32 GB recommended
CPU	Modern 6-core minimum; 8-core+ recommended
Storage	At least 100 GB free; 250–500 GB recommended
OS	Windows 10 22H2+ / Windows 11

3. Core Architecture



4. Why These Two Models

Qwen3-8B is used as the primary model because its model card describes strong instruction following, reasoning, coding, multilingual support, and tool/agent capabilities. It supports both thinking and non-thinking modes. DeepSeek-R1 provides reasoning-oriented distilled models, including an 8B Llama-based option; the DeepSeek project also publishes Qwen-based distilled variants. For an RTX 4060 8 GB, an 8B-class quantized deployment is the practical starting point.

Official references: Qwen3-8B model card and DeepSeek-R1 model card.

5. Software Stack

Recommended stack: Windows + NVIDIA driver + Ollama + Python 3.12 + FastAPI + HTTP client + a web-search provider + optional vector database for RAG.

Ollama runs natively on Windows, supports NVIDIA GPUs, and exposes a local API at <http://localhost:11434>. It can be used from Python or JavaScript.

6. Install Ollama

Install Ollama for Windows from the official Ollama website. After installation, verify the command:

```
ollama --version
nvidia-smi
```

Ollama's Windows documentation states that the application runs natively and that its API is available at localhost:11434.

7. Download the Models

```
ollama pull qwen3:8b
ollama pull deepseek-r1:8b

ollama list
```

Model tags can change as Ollama's model library evolves. If the exact DeepSeek tag is unavailable, use the currently listed 8B DeepSeek-R1 distilled model/tag and keep the application model name configurable in an environment file.

8. Test Each Model

```
ollama run qwen3:8b
# Ask: Explain TCP three-way handshake in simple English.

ollama run deepseek-r1:8b
# Ask: Verify whether TCP is connection-oriented and explain why.
```

9. Project Folder

```
AI-Assistant/
├── backend/
│   ├── main.py
│   ├── router.py
│   ├── ollama_client.py
│   ├── qwen.py
│   ├── verifier.py
│   ├── web_search.py
│   ├── rag.py
│   └── prompts.py
├── data/
│   └── documents/
├── frontend/
├── .env
├── requirements.txt
└── README.md
```

10. Python Environment

```
python -m venv .venv
.venv\Scripts\activate

pip install fastapi uvicorn httpx python-dotenv pydantic
```

11. Ollama API Design

The application should call Ollama locally rather than invoking the model through a remote API. The default local API base is `http://localhost:11434/api`.

```
POST http://localhost:11434/api/chat

{
  "model": "qwen3:8b",
  "messages": [
    {"role": "system", "content": "You are the primary assistant."},
    {"role": "user", "content": "Explain TCP."}
  ],
  "stream": false
}
```

12. Qwen Draft Prompt

You are the primary AI assistant.

Answer the user's question clearly and accurately.
Use the supplied reference information when present.
For current information, rely on the supplied sources rather than
inventing facts.
Write natural, grammatically correct English.
Do not mention internal model processing.

USER QUESTION:
{question}

REFERENCE DATA:
{context}

13. DeepSeek Verification Prompt

You are a strict answer verifier.

USER QUESTION:
{question}

QWEN DRAFT:
{draft}

REFERENCE INFORMATION:
{context}

Check:

1. Relevance
2. Factual accuracy
3. Correct use of current information
4. Unsupported claims
5. Contradictions
6. English clarity
7. Missing important information

Return:
VERDICT: CORRECT or INCORRECT

If INCORRECT, list the errors and provide corrected facts.
If CORRECT, briefly explain why the draft is acceptable.

14. Correction Prompt

Rewrite the answer using the verifier's corrections.

USER QUESTION:
{question}

ORIGINAL DRAFT:
{draft}

VERIFIER FEEDBACK:
{feedback}

REFERENCE INFORMATION:
{context}

Rules:

- Preserve correct information.
- Fix every identified factual error.
- Do not add unsupported facts.
- Answer the user's actual question directly.
- Use clear, natural English.
- Do not mention the verifier or internal processing.

15. Current Information / Web Search

A local model's weights are not a live news feed. Add a search layer for queries containing signals such as latest, today, current, recent, this week, price, release, version, news, update, or other time-sensitive intent. The search layer should retrieve a small set of relevant sources, preserve title/date/URL/snippet, and pass the evidence to Qwen. DeepSeek then verifies whether the draft actually reflects the retrieved evidence.

```
USER QUESTION
|
v
CURRENT-INTENT DETECTOR
|
+---- no ----> Qwen
|
yes
|
v
WEB SEARCH
|
v
SOURCE FILTER
|
v
Qwen + sources
|
v
DeepSeek verification
|
v
Final answer + sources
```

16. Source Quality Rules

- Prefer official documentation, government sources, primary announcements, and reputable reporting.
- Keep publication/update dates where available.
- Do not treat a search snippet alone as definitive evidence for high-stakes claims.
- If sources disagree, show the disagreement or retrieve additional primary sources.
- Do not claim that information is current unless the retrieved source is recent enough for the question.
- Include source links in the final UI for current-information answers.

17. RAG Extension

For your own PDFs and notes, add retrieval-augmented generation. Extract documents, split them into chunks, create embeddings, store them in a vector database, retrieve the most relevant chunks, and pass those chunks to Qwen.

DeepSeek then verifies the answer against the retrieved document context.

18. Sequential Model Execution

Because the RTX 4060 has 8 GB VRAM, do not design the first version around simultaneously resident 8B models. Execute the pipeline sequentially: Qwen generates the draft, then DeepSeek verifies it, then Qwen is called again only when correction is needed. This reduces peak GPU pressure and avoids unnecessarily running both models at once.

19. Backend Flow

```
async def answer_question(question):
    intent = classify_intent(question)

    context = ""
    if intent.needs_web:
        context = await web_search(question)

    if intent.needs_rag:
        context += await retrieve_documents(question)

    draft = await qwen_generate(question, context)

    verdict = await deepseek_verify(
        question=question,
        draft=draft,
        context=context
    )

    if verdict.is_correct:
        return finalize(draft, context)

    corrected = await qwen_correct(
        question=question,
        draft=draft,
        feedback=verdict.feedback,
        context=context
    )

    return finalize(corrected, context)
```

20. API Endpoints

```
POST /chat
POST /search
POST /documents/upload
GET /health
GET /models
```

21. Recommended Development Order

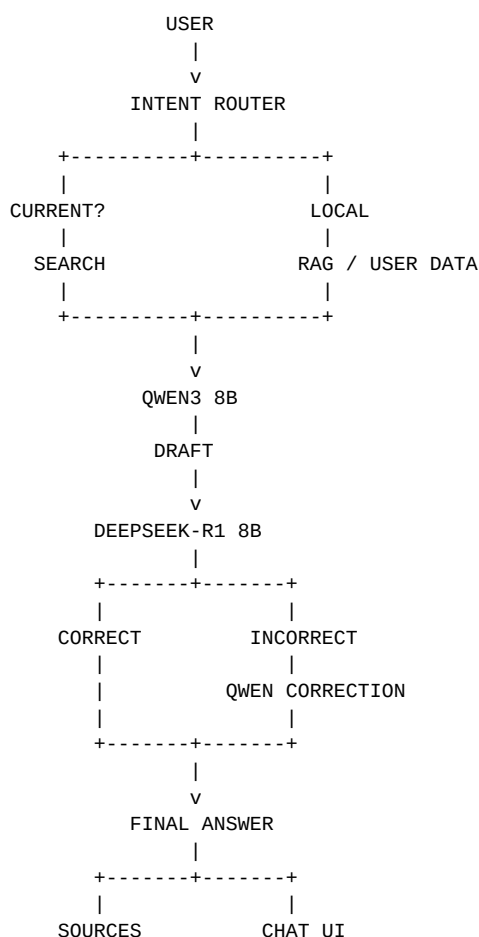
1. Install NVIDIA driver and confirm nvidia-smi.
2. Install Ollama and confirm its local API.
3. Download and test Qwen3 8B.
4. Download and test DeepSeek-R1 8B.
5. Build the Qwen draft endpoint.
6. Build the DeepSeek verification endpoint.
7. Implement CORRECT/INCORRECT branching.
8. Implement Qwen correction.
9. Add the current-information router.
10. Add web search and source formatting.

11. Add RAG for personal/college documents.
12. Build the chat UI.
13. Add logging, latency measurements, error handling, and source display.

22. Testing Plan

Test type	Example
General	What is TCP?
English	Correct this sentence and explain the correction.
Reasoning	Solve a multi-step logic or mathematics problem.
Current	What are the latest developments in [topic]?
Contradiction	Provide a question where a supplied source conflicts with general model knowledge.
RAG	Ask a question answered only by an uploaded PDF.

23. Final System



24. Official References

Ollama Windows documentation: <https://docs.ollama.com/windows>

Ollama API documentation: <https://docs.ollama.com/api>

Qwen3-8B model card: <https://huggingface.co/Qwen/Qwen3-8B>

DeepSeek-R1 model card: <https://huggingface.co/deepseek-ai/DeepSeek-R1>

Note: Model tags, Ollama behavior, NVIDIA driver requirements, and available search APIs can change. Verify current versions before deployment. The architecture is intentionally modular so the model names and search provider can be changed without redesigning the application.